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# Telecom Churn Prediction & Proactive Retention Proposal [Company A]

## 1. Market Analysis

The global wireless telecom market exceeded **USD 1.7 trillion** in 2024. Key dynamics:

- **Saturation**: smartphone penetration >80% in developed markets; growth depends on ARPU and stealing market share from competitors.
- **Churn cost**: acquiring a new subscriber costs **5–25× more** than retaining one. Average annual churn in US telecom: 15–25%.
- **5G transition pressure**: customers with older handsets are prime targets for competitor upgrade deals.
- **Price sensitivity**: overage charges on voice and data are a leading driver of switching.

**Business problem**: Company A has no ML capability and ~50% annual churn — losing roughly one in two customers per year. The core question is: *which customers are about to leave, and what can we do to keep them?*

## 2. Data Overview

The dataset contains historical data from Company A split across two files:

| File | Rows | Columns | Content |
|---|---|---|---|
| `Client.csv` | 100,000 | 50 | Account info: tenure, demographics, equipment |
| `Record.csv` | 100,000 | 51 | Usage, billing, call quality + churn label |

| File | Rows | Columns | Content |
|---|---|---|---|
| **Merged** | **100,000** | **100** | Joined on `Customer_ID` (1-to-1) |

# 3. Exploratory Data Analysis

## Churn distribution

| Class | Count | % |
|---|---|---|
| Stayed (0) | 50,438 | 50.4% |
| Churned (1) | 49,562 | 49.6% |

The dataset is nearly balanced — **accuracy is a valid metric**. Real-world telecom churn is typically 5–10%, so the 50% rate signals a critical retention crisis at Company A.

## Key findings

**Equipment age (`eqpdays`)**: Churners have significantly older devices. Customers with equipment older than 2 years are disproportionately likely to churn — likely switching to a competitor for an upgrade deal.

**Revenue & usage trends (`change_rev`, `change_mou`)**: Churners show declining revenue and minutes-of-use vs. their 3-month average. This trajectory is detectable months before cancellation.

**Customer care calls (`custcare_Mean`)**: Churners place more customer care calls on average. High complaint volume is an early warning signal and an opportunity to intervene.

**Call quality (`drop_vce_Mean`, `blck_vce_Mean`)**: Churners experience slightly more dropped and blocked calls, suggesting network quality contributes to dissatisfaction.

**Tenure (`months`)**: Shorter-tenured customers churn at higher rates, consistent with industry patterns where early-lifecycle customers are less sticky.

# 4. Problem Definition

- **ML task**: Binary classification — predict whether a customer will churn (`churn = 1`) in the next 31–60 days.
- **Target variable**: `churn`

- **Primary metric**: ROC-AUC (threshold-independent, captures ranking quality)
- **Secondary metrics**: Accuracy, F1-score

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## 5. Data Preparation & Feature Engineering

**Preprocessing steps:**

- Dropped columns with >50% missing values (`numbcars`, `dwllsize`, `HHstatin`, `ownrent`, `dwlltype`)
- Label-encoded all categorical columns
- Imputed remaining missing values with column median
- 80/20 stratified train/test split (`random_state=42`)

**Engineered features:**

| Feature | Definition | Rationale |
|---|---|---|
| `completion_rate` | `comp_vce_Mean / (plcd_vce_Mean + 1)` | Fraction of placed calls that completed |
| `overage_ratio` | `ovrrev_Mean / (rev_Mean + 1)` | Overage as share of total revenue |
| `rev_trending_up` | `change_rev > 0` | Is revenue growing? |
| `mou_trending_up` | `change_mou > 0` | Is usage growing? |
| `eqpdays_bucket` | Cut into <6mo / 6-12mo / 1-2yr / >2yr | Equipment age segment |

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## 6. Models & Results

Three models trained and evaluated on the held-out test set (20,000 customers):

| Model | ROC-AUC | Accuracy | F1 |
|---|---|---|---|
| Logistic Regression | [fill after running] | [fill] | [fill] |
| Random Forest | [fill after running] | [fill] | [fill] |
| **XGBoost** | **[fill after running]** | **[fill]** | **[fill]** |

**Why XGBoost?**

- Handles non-linear relationships and feature interactions naturally
- Robust to outliers common in telecom billing data
- Feature importance scores give direct business interpretability

**Top predictive features** (from XGBoost): `eqpdays`, `months`, `avgrev`, `totrev`, `mou_Mean` — consistent with EDA findings.

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# 7. Business Proposal

## Solution: Proactive Retention Scoring System

Run the XGBoost model monthly. Every customer gets a churn probability score (0–1), enabling tiered interventions:

| Risk Tier | Score | Action |
|---|---|---|
| High Risk | ≥ 0.75 | Personal retention call + subsidized handset upgrade offer |
| Medium Risk | 0.50 – 0.75 | Automated SMS/email with loyalty discount or data bonus |
| Low Risk | < 0.50 | No intervention — preserve margin |

## Estimated impact (conservative assumptions)

| Metric | Value |
|---|---|
| Active customers | 100,000 |
| Monthly churn rate (~4%) | ~4,000 customers/month |
| Avg. monthly revenue per customer | ~$50 |
| High-risk customers per month (≥0.75) | ~8,000 |
| Customers saved (15% retention rate) | ~1,200/month |
| Monthly revenue retained | ~$60,000 |
| Annual revenue retained | ~$720,000 |
| Intervention cost ($10/contact × 8,000) | $80,000/month |
| **Net annual benefit** | **~$640,000** |

## Key action levers

1. **Equipment upgrade program**: Proactively offer subsidized upgrades to customers with `eqpdays > 730` AND churn score > 0.6.
2. **Usage decline alert**: Trigger outreach when `change_mou` or `change_rev` drops >20% vs. 3-month average.
3. **Customer care fast-track**: Assign dedicated account managers to high-complaint customers before they cancel.

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## 8. References


1. Statista (2024). *Global telecom services market size worldwide.* https://www.statista.com/statistics/1142372/
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